

ICT SYLLABUS FOR UPPER SECONDARY (ALL PATHWAYS)

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FOREWORD

The Rwanda Basic Education Board (REB) is honoured to present the second edition of the Upper Secondary School ICT syllabus (S4–S6). It serves as an official guide for competence-based teaching and learning, ensuring consistency in delivering quality education across the country. Developed collaboratively by subject specialists, educators, and curriculum developers, it supports the implementation of the Competence-Based Curriculum (CBC) and reflects Rwanda’s commitment to nurturing learners’ knowledge, skills, values, and attitudes.

Quality education is the foundation of Rwanda’s sustainable development and prosperity. Aligned with the national education philosophy, this syllabus promotes critical thinking, creativity, inclusiveness, and holistic learner development. It aims to empower young people with competencies necessary to achieve full potential, equipping them with relevant knowledge, skills, the right attitudes and values to integrate effectively into society and pursue opportunities in a competitive world.

Anchored in Rwanda Vision 2050 and the National Strategy for Transformation (NST2), this syllabus supports the nation’s goal of building a knowledge-based economy driven by innovation, resilience, and global competitiveness. It prioritizes learner-centred pedagogy, active learning, and the practical application of knowledge to prepare student not only for academic success but also to become responsible, ethical, and productive citizens capable of contributing to Rwanda’s continued transformation.

This syllabus reflects our determination to equip learners with 21st-century skills, foster innovation, and instill values of integrity, social responsibility, and lifelong learning. Its success depends largely on the dedication and professionalism of teachers and school leaders. We therefore call upon all educators to embrace the approaches, methodologies, and assessment strategies outlined herein with commitment and passion, transforming classrooms into dynamic spaces of curiosity, creativity, and inspiration.

On behalf of REB, I extend my heartfelt appreciation to all who contributed to the development of this syllabus. Your expertise and dedication have been instrumental in shaping a curriculum that responds to the needs of our learners and supports Rwanda’s national aspirations. Together, we can ensure that education continues to be a powerful driver of our nation’s development and prosperity.

Dr. MBARUSHIMANA Nelson
Director General, REB



ACKNOWLEDGEMENT

The development of this syllabus has been a collective effort that reflects the dedication, expertise, and commitment of many stakeholders. On behalf of the Curriculum, Teaching and Learning Resources Department of the Rwanda Basic Education Board (REB), I wish to express my deepest appreciation to all who contributed to the successful preparation of this Upper Secondary school ICT syllabus (S4–S6).

Special thanks go to the subject experts, curriculum developers, teachers, and education partners whose insights, feedback, and professional contributions have been invaluable in shaping this syllabus. Your collaboration has ensured that the syllabus is not only aligned with Rwanda's Competence-Based Curriculum but also responsive to the aspirations of Vision 2050 and the National Strategy for Transformation (NST2).

We also acknowledge the invaluable support of development partners and education stakeholders who have consistently provided technical assistance in curriculum adaptation. Your partnership has greatly contributed to strengthening the quality of Rwanda's education system.

Finally, we call upon teachers, school leaders, and all education practitioners to use this syllabus with commitment and professionalism, ensuring that classrooms become vibrant learning environments that nurture knowledge, skills, values, and positive attitudes among learners. Through your dedication, this syllabus will fulfil its purpose of empowering learners to realize their full potential and contribute meaningfully to Rwanda's sustainable development.



MURUNGI Joan

Head of Curriculum, Teaching and Learning Resources Department

Rwanda Basic Education Board (REB)

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1. GENERAL INTRODUCTION

1.1. Background to the syllabus adaptation

The adaptation of the ICT syllabus for upper secondary school reflects Rwanda's ongoing commitment to strengthening the Competence-Based Curriculum (CBC), first rolled out in 2015. Continuous feedback from teachers, school leaders, and education partners highlighted the need for improvements in content delivery, time allocation, and responsiveness to learner needs.

This second edition incorporates the Ministerial directive to harmonize school hours to five per day and 33 weeks of learning annually and ensures greater coherence, inclusiveness, and alignment with national priorities such as *Vision 2050* and the *National Strategy for Transformation (NST2)*, which emphasize a knowledge-based economy, responsible citizenship, and sustainable development.

ICT at the upper secondary level aims to equip learners with foundational knowledge, practical skills, and positive values needed to identify opportunities, solve real-life problems, and contribute meaningfully to their communities. The subject encourages learners to think creatively, act ethically, and develop a sense of initiative, self-confidence, and social responsibility.

2. TEACHING AND LEARNING ICT

2.1. Rationale of teaching and learning ICT

Information and Communication Technology is a pillar to Rwanda's transformation into a knowledge based economy. ICT is penetrating every aspect of every day's life including service delivery and the country is going towards an intensive integration of ICT in education, which is a key player in the development of a country. ICT in education is to be used to enhance teaching and learning and prepare citizens who will adapt in this newly created environment. New technologies are rapidly evolving requiring citizens to quickly adapt in order to be able to have access to services and integrate in the global World.

Students in upper secondary have therefore a sound reason to study ICT topics related to Computer System, Computer Programming, Database and Emerging Technologies. Students will explore practical applications of ICT through hands-on activities, simulations, and collaborative projects, building confidence and competence in digital environments.

2.1.1. ICT and Society

Information and Communications Technology (ICT) is a dominant aspect of modern life that has changed every aspect of human society. The government focuses on the comprehensive use of ICT to reach its goals stipulated in different official documents and policies. In Rwandan society ICT is the foundation for long term, sustainable and efficient government services, GDP growth, communication, economic development, financial transactions, and the global distribution of jobs. Advances in information influence the value of investments, living standards, patterns of work and leisure, business efficiency and productivity, and spur competition in the market place.

2.1.2. ICT and the Student

ICT is a subject that empowers students with a package of skills through the use of various ICT tools namely computers, the internet and new technologies such as AI in providing and managing information.

ICT as a subject provides basic computing skills and exposes learners to common Google tools such as Google Docs, Google Sheets and Google Slides while it introduced these students to AI topics which is now projected to be the new future. ICT also opens the students to the world of programming in order to initiate them at an early age to the logic of program creation which they will later embrace and pioneer this field in Rwanda and the World.

2.2. Competences

A competence is the ability to use an appropriate combination of knowledge, skills, attitudes, values, and behaviours to accomplish a task successfully. In this syllabus, competences reflect a holistic blend of these elements, enabling learners to apply them confidently and appropriately in diverse real-life situations.

By acquiring these competences, learners become better equipped to handle challenges, make informed decisions, act responsibly, and contribute positively as active members of society. The competences are embedded in the broad subject competences, yearly objectives,

and unit-specific learning competencies. Both generic competences and basic competences must be emphasized and consistently reflected in the teaching and learning process.

Teachers are encouraged to design learning experiences that engage learners in meaningful tasks, allowing them to practice and demonstrate competences both during and at the end of the learning process.

2.2.1. Generic competences

Generic competences are cross-cutting abilities developed across all subjects. They equip learners with essential lifelong learning skills and underpin all other competences in the curriculum.

- **Critical and problem-solving skills:** Learners develop the ability to think broadly and objectively, evaluate their causes and consequences, and suggest peaceful and practical solutions that contribute to harmony and sustainable living in their communities.
- **Creativity and innovation:** The acquisition of such skills will help learners to take the initiative and use imagination beyond the knowledge provided, to generate new ideas and construct new concepts.
- **Research:** This will help learners acquire the ability to gather, analyze, and interpret information from a variety of sources, including observations, interviews, and case studies to inform decisions and develop evidence-based conclusions.
- **Communication in official languages:** Teachers, irrespective of their subject, will ensure proper use of the language of instruction to help learners communicate clearly, confidently, and respectfully in both spoken and written forms, enabling them to express ideas, negotiate, and present information effectively for logical and coherent exchange.
- **Cooperation, interpersonal management and life skills:** This competence enables learners to work effectively with others, respect diverse opinions, and show empathy in group activities. It fosters moral values, responsibility, and active participation in community-oriented tasks such as environmental protection, health promotion, and service. Learners also build social skills that enhance teamwork, leadership, and adaptability in addressing life's challenges.

- **Lifelong learning:** Learners cultivate curiosity, initiative, and adaptability by linking classroom knowledge to real-life experiences. This empowers them with the ability to update knowledge and skills independently, cope with evolving societal needs, and pursue personal growth and continuous development.

2.2.2. ICT broad competences

At the end of upper secondary education, learners should be able to demonstrate the following broad competences specific to ICT:

- Organise, manage, and manipulate information effectively by using widely available digital tools,
- Apply computational thinking, logical and algorithmic precision in problem solving and creativity,
- Demonstrate responsible and ethical use of computers, both online and offline, for personal, academic, and professional contexts, while showing awareness of key ethical issues in ICT,
- Apply all stages of software development life cycle (SDLC) in the process of designing and implementing software solutions,
- Analyze a computer program so as to identify errors and correct them by applying systematic principles of debugging,
- Design and develop small-scale software applications that address specific user needs and community challenges,
- Design, configure, and explain the functioning of traditional computer networks, Internet of Things (IoT) systems and robotic system,
- Demonstrate advanced understanding of ICT learned topics so as to communicate this knowledge effectively to diverse audiences, including both ICT-literate individuals and those with limited or no ICT background,
- Use artificial intelligence (AI) solutions productively and responsibly to support learning, work, and community development,
- Analyse the challenges faced by communities, propose appropriate AI solutions that can contribute to alleviating them, design those solutions and develop small-scale AI projects.

2.2.3. ICT and developing competences

National policies emphasize developing competences that promote critical thinking and responsible citizenship through interactive and practical learning.

ICT contributes to a learner's development of critical thinking, research and problem solving, creativity and innovation, communication and cooperation skills. This will be achieved by conceptualising, analyzing, synthesising, evaluating, manipulating and applying the role of various computing tools to appropriate contexts of real life, as well as analysing both hardware and software. Learners will be able to use ICT tools safely, securely and ethically to manage computing tools, organise and present information using different software

Through ICT, learners will identify problems in information processing and conduct analysis by considering relevant functional, practical, human and socioeconomic factors. They will also draw up specifications for the ICT enabled solutions to problems. Learners will draw on a range of resources which are most suitable for solving problems and developing solutions and use appropriate methods to implement these solutions using equipment, tools and techniques sensibly. They will test, evaluate and refine solutions systematically, document solutions to problems, and appropriately apply ICTs to solve problems

3. PEDAGOGICAL APPROACHES

The pedagogical approach in ICT under the CBC emphasizes learner-centered, participatory, and inclusive methods that develop knowledge, skills, values, and competences for responsible citizenship. Teaching is interactive and active, engaging learners through debates, discussions, storytelling, field visits, projects, research, case studies, and reflective activities. These approaches ensure that learning is enjoyable, personalized, and connected to real-life situations, while also fostering critical thinking, moral reasoning, and problem-solving.

3.1. Role of the teacher

The teaching and learning of ICT at upper secondary level must reflect the Competence-Based Curriculum (CBC) principles, which promote learner-centred, participatory, and practical methods. The main goal is to guide learners acquire and apply knowledge, skills, values, and attitudes that prepare them to become innovative, responsible, and productive citizens.

Teachers should therefore move beyond traditional lecturing to interactive strategies that encourage inquiry, creativity, problem-solving, and collaboration. Learning must link theory with practice, making classroom experiences relevant to the learner's environment and everyday life.

The teacher must identify the needs of learners, the nature of the learning to be done, and the means to shape learning experiences accordingly. A teacher's role is to organise the learners in the classroom or computer laboratory and other ICTs to engage them in the use of these ICTs through participatory and interactive methods. This ensures that the learning is personalised, active, participative and

cooperative. The teacher must devise remedial strategies to ensure low achievers and those with learning difficulties keep pace with the rest in acquiring the required competencies.

3.2. Role of the learner

Learners are active participants and co-creators of knowledge. They engage in inquiry, debates, problem-solving, and group collaboration while drawing connections between personal experiences, social realities, and religious teachings. Through these activities, learners develop critical thinking, communication, respect for diversity, moral responsibility, and the ability to apply knowledge in everyday life and societal challenges.

3.3. Inclusive and Special Needs Education

In line with Rwanda's inclusive education policy, all learners regardless of their abilities, have the right to access quality education. Teachers adapt teaching methods, materials, and assessment strategies to meet diverse needs, including those of learners with disabilities. Differentiated instruction, assistive technologies, peer support, and individualized approaches are used to ensure equitable participation and achievement of learning outcomes. This inclusive approach recognizes learners' varied paces, experiences, and abilities, creating an environment where every learner can thrive.

4. ASSESSMENT APPROACHES

Assessment in the Competence-Based Curriculum (CBC) is an essential part of the teaching and learning process, designed to evaluate learners' progress in knowledge, skills, values, and attitudes while supporting continuous growth. It emphasizes real-life application and problem-solving, enabling learners to demonstrate their competences in authentic contexts. Assessment involves systematically gathering and interpreting evidence to measure performance against established standards, providing constructive feedback to guide both teaching and learning.

In ICT, assessment focuses on learners' ability to demonstrate creativity, innovation, problem-solving, ethical behaviour, and teamwork. It encourages reflection, continuous improvement, and the development of responsible entrepreneurial mindsets.

Assessment should therefore be comprehensive, covering both formative (continuous) and summative (end-of-term or national) evaluations. It should be transparent, fair, inclusive, and competency-aligned.

4.1 Levels of assessment in Advanced Level across terms and national examinations

To provide a comprehensive picture of learner achievement and development, assessment is conducted at multiple levels:

- (i) **Diagnostic:** Conducted before instruction to identify learners' prior knowledge and skills; administered at the beginning of a unit or lesson to guide teaching.
- (ii) **Formative / Continuous:** Carried out during the learning process to track progress and provide feedback; implemented continuously through quizzes, projects, class activities, and teacher observations.
- (iii) **Summative assessment:**
 - **End Term I:** Administered after instruction to evaluate learners' achievement of Term I content; conducted at the end of the first term as a school-based assessment.
 - **End Term II:** District-coordinated assessment designed to measure achievement of Term II content; developed by schools and quality-checked at district level.
 - **End Term III:** National assessment to evaluate mastery of the entire year's content; prepared, set, and moderated by NESa at the end of the third term.
 - **National Examinations (NESa):** High-stakes assessments for certification, selection, and accountability; administered nationally at the end of key education cycles (P6, S3, S6) under NESa's authority. This examination integrates objective questions, structured responses, and applied tasks aligned with CBC principles to holistically measure learner achievement.
 - In Advanced Level, all students will take **11 subjects** in total -**7 core subjects** and **4 compulsory subjects**. The compulsory subjects include ICT, Entrepreneurship, **General and Religious Studies**, and Physical Education and Sport, which are mandatory for all learners across all pathways.
 - All subjects will be **assessed at the national level**, except for **Physical Education and Sport, General and Religious Studies, ICT**, and **Entrepreneurship**, which will be **assessed at the school level only**.

Average scores from school, district, and comprehensive summative assessments will be weighted and included in the final national examination grading, initially contributing 10%, with the percentage gradually increasing as teachers gain experience and confidence in assessment practices.

4.2. Record keeping

Accurate record keeping is essential for monitoring learner progress and informing instructional decisions. Teachers use tools such as portfolios, checklists, and report books to collect evidence of learning. Portfolios include samples of learners' work, test results, projects, and self-reflections, serving as both a record of achievement and a diagnostic tool for guiding remedial action. These records provide data for feedback to learners, teachers, and parents, and contribute to final evaluations.

4.3. Item writing in summative assessment

Summative assessment items must be valid, fair, reliable, and aligned with CBC principles. They should test a range of competencies, including knowledge recall, application, analysis, and values integration, while reflecting real-life contexts. Item writers follow a structured process that involves:

- Identify topic areas to be tested from the subject syllabus.
- Outline subject matter content to be considered as the basis for the test.
- Identify learning outcomes to be measured by the test.
- Preparing a test blueprint or table of specifications;
- Using verbs that require higher-order thinking beyond memorization;
- Applying transparent and consistent marking schemes.
- Greater weight is given to higher-order cognitive skills as stated in the syllabus.

4.4. Structure and format of the examination

At the completion of upper secondary school education, learners undertake a national examination designed to evaluate literacy, numeracy, and competences in core subjects. At the time of review of this curriculum, ICT is not included as a subject in national examinations assessment. However, it is projected that the scores of learners at school level may contribute to the national exams' results to a certain percentage.

4.5. Reporting to parents

Regular communication with parents is an essential component of the CBC. This is achieved through progress reports, parent–teacher meetings, and school community activities. Since the CBC emphasizes a wide range of learning outcomes: knowledge, skills, attitudes, and competences a single mark is insufficient to capture learner progress. Effective reporting should highlight areas where learners are excelling as well as those requiring improvement. Using descriptive feedback or simple rating scales (e.g., meeting expectations very well,

meeting expectations, not meeting expectations) across knowledge, subject-specific skills, and competences provides a clearer and more comprehensive picture than numerical grades alone. This approach fosters meaningful parental involvement and supports learners' continued growth.

5. RESOURCES

The effective implementation of the **ICT** syllabus requires the provision and utilization of appropriate resources. These resources include both teaching and learning materials and qualified human personnel, without which the delivery of the syllabus would be ineffective.

Teachers are encouraged to utilize both locally available and digital resources to create rich, relevant, and engaging learning experiences that promote creativity, problem-solving and innovation.

5.1. Materials needed for implementation

A variety of materials are essential to facilitate teaching and learning processes in ICT. These materials provide learners with practical experiences, stimulate critical thinking, and foster the acquisition of competences. They include:

- **Textbooks and reference materials:** Updated textbooks, teacher's guides, learner's books, dictionaries, and business-related reference materials that support theoretical and practical learning.
- **Teaching and learning aids:** The ICT subject will be taught mostly in computer laboratories using computers preferably with internet connectivity, projectors, tablets, smart phones or any other material that the teacher will judge necessary. Visual and audio-visual resources such as charts, posters, infographics, diagrams, models, and videos, as well as digital tools like e-books, educational websites, and online platforms, can be used to enhance learning.
- **Library and resource centre materials:** Access to a well-equipped library with supplementary reading materials to extend the knowledge beyond what is got from the classroom.
- **Simulation resources:** For better understanding of different topics such as robotics and IoT, the teacher may use simulations that are available on the internet by giving a preference to those that are downloaded and installed on the computer.
- **Field visit resources:** Materials to support educational tours and fieldwork, such as journals, notebooks, and logistical support, since field visits and community interactions are integral to experiential learning in this subject. This will make learners discover how the different ICT topics are important in real life.
- **Assessment tools:** Rubrics, checklists, portfolios, and test materials are necessary to support continuous and summative assessment in line with the competence-based approach.

These resources should be adapted to the learners' context, ensuring inclusivity and accessibility for learners with special educational needs.

5.2. Human resource

The effective implementation of this curriculum requires a joint collaboration of educators at all levels. Given the material requirements, teachers are expected to accomplish their noble role. School head teachers and directors of studies are required to follow-up and assess the teaching and learning of ICT. These combined efforts will ensure bright future careers and lives for learners as well as the contemporary development of the country.

In a special way, a teacher of ICT at ordinary level should have a firm understanding of ICT concepts at the level he/she teaches. He/she should be qualified in **ICT or Computer Science related fields** with Education and have firm ethical conduct. The teacher should possess the qualities of a good facilitator, organiser, problem solver, listener and adviser. He/she is required to have basic skills and competency of guidance and counseling because students may come to him/her for advice.

5.3. Skills required for the teacher of ICT

Effective implementation of this Syllabus requires teachers with strong subject knowledge, pedagogical skills, and interpersonal competence. As facilitators and role models, they guide learners to develop competences in ICT and apply them in real-life contexts. To achieve this, teachers should demonstrate the following key skills:

a) **Subject matter expertise:**

- Teacher should not only understand the different topics in the ICT subject but also extend their knowledge beyond them so that they become able to teach.
- Capacity to adapt to the fast evolution of technology, including the emergence and widespread adoption of innovations such as artificial intelligence, robotics and IoT within the community. The teacher of ICT should be a pioneer in these technologies and help spread them to his students and then to the community.
- Understand national aspirations as stated in different policies and relate the role of ICT as a subject in achieving them.
- Link the use of ICT with other subjects and real life situations.

b) **Pedagogical and instructional skills**

- Ability to design and implement learner-centred and participatory lessons that encourage active engagement, creativity, and critical thinking.
- Use of diverse teaching methods such as brainstorming, project-based learning, role-playing, case studies, and community engagement.
- Creativity and innovation in developing and applying diverse teaching resources, including ICT tools and community-based materials.
- Ability to adjust instruction to learners' levels, ensuring inclusivity and engagement for all.

c) Assessment skills

- Proficiency in designing and applying formative and summative assessments aligned with curriculum competencies.
- Effective use of rubrics, portfolios, checklists, and peer assessments to evaluate learners' competences and provide feedback that motivates learners and guides improvement.
- Skill in developing valid and fair test items that assess higher-order thinking and practical skills.
- Ability to analyze assessment results and use them to guide remedial or enrichment activities.

d) Classroom management and facilitation

- Ability to establish a respectful, safe, and inclusive learning environment that promotes tolerance and values diversity.
- Proficiency in managing group work, discussions, and collaborative learning activities.
- Consistent modelling of positive behaviour, respect, punctuality, and integrity as an example for learners.

e) Communication and interpersonal

- Strong oral and written communication to simplify complex ideas, stimulate dialogue, and foster critical reflection.
- Active listening and empathy to respond effectively to learners' academic and personal needs.
- Collaboration with colleagues, parents, school leaders, and community stakeholders to support holistic learner growth.
- Ability to motivate and inspire learners through encouragement and positive reinforcement.

f) Guidance and counseling skills

- Ability to provide moral, emotional, and motivational guidance to learners to build confidence, resilience, and ethical conduct.
- Demonstrating passion, patience, impartiality, and genuine care in dealing with learners' individual differences and challenges.
- Supporting learners in exploring business ideas, career paths, and responsible life choices.

g) **Professional and ethical competence**

- Upholding high ethical standards, integrity, impartiality in all teaching and assessment practices, and ethical conduct, serving as a role model for learners,
- Capacity to balance academic instruction with the cultivation of values and responsible citizenship,
- Demonstrating commitment to continuous professional development (CPD) through reflective practice, peer collaboration, and research.

h) **Research and innovation skills**

- Engagement in reflective practice and classroom-based research to improve teaching and learning.
- Adaptation of innovative approaches and new technologies to enhance learner participation.
- Encouraging learners to use research findings in designing projects and solving community challenges.

6. SYLLABUS UNITS

6.1. Presentation of the structure of the syllabus

The ICT syllabus is designed as a single, integrated syllabus implemented uniformly across all grades. Its structure follows a logical progression that ensures clarity, consistency, and alignment with the Competence-Based Curriculum approach.

At each grade level, the syllabus is organized into **Topic Areas**, which are further broken down into **Sub-Topic Areas**. Each Sub-Topic Area is composed of **Units**, and every unit is characterized by the following features:

1. **Alignment with Lessons:** Each unit corresponds to a specified number of lessons, ensuring effective time allocation and instructional planning.
2. **Key Unit Competence:** Every unit is built around a *Key Unit Competence* that guides all teaching and learning activities. This competence represents the overall learning goal to be achieved by both teacher and learners.
3. **Learning Objectives:** Each key unit competence is translated into specific learning objectives, categorized as follows:
 - **Type I (Knowledge and Understanding):** Focused on foundational knowledge and comprehension, often referred to as *Lower Order Thinking Skills*.

- **Type II and Type III (Skills, Attitudes, and Values):** Aimed at developing practical skills, critical thinking, ethical values, and positive attitudes. These objectives, known as *Higher Order Thinking Skills (HOTS)*, form the core focus of the adapted Competence-Based Curriculum.
4. **Content Specification:** Each unit outlines the scope of content to be taught and learned, ensuring that teachers and learners work within the boundaries set by the stated learning objectives.

6.2. Senior Four

6.2.1. General competences at the end of Senior 4:

- Create documents using Google Docs to enhance communication, organization, and teamwork in a shared working environment.
- Design an optimized digital logic circuit implementing Boolean logic
- Develop algorithmic solutions from problems using pseudocode and flowcharts,
- Create Python programs that involve decision making and operations on data,
- Develop a well-structured static website
- Apply styles in web pages for better visual presentation and usability,
- Design a normalized relational model using entity-relationship diagrams and appropriate keys,

6.2.2. ICT Units for Secondary Four

Topic Area: Computer System			Sub-topic area: Computer software	
S4 ICT		Unit 1: Google Docs		Nº of periods: 8
Key Unit Competence: Create documents using Google Docs to enhance communication, organization, and teamwork in a shared working environment.				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
• Explain the use of Google Docs • Describe how to access Google Docs through Gmail or Google Drive. • Identify parts of the Google Docs interface	• Create a new document • Navigate the Google Docs interface • Apply the basic text formatting • Insert images,	• Appreciate the importance of collaboration and teamwork in digital environments. • Develop a sense of responsibility when sharing and editing documents online.	Introduction to Google Docs: • Importance of Google Docs • Accessing Google Docs (via Gmail/Drive) • Navigating the Google Docs interface (menu, toolbar, workspace) • Creating a new document Text formatting basics • Changing font style, size, and color • Bold, italics, underline, and highlighting • Alignments (left, center, right, justify)	• Group discussion about the use of Google Docs • Individually learners follow the teacher’s demonstration to create and format their own document. • In groups students collaborate on a shared document to apply

• State the steps for creating a new document. • Explain basic text formatting features and their functions. • Describe how to create and format lists and tables. • Identify the process of inserting elements • Explain sharing and collaboration features in Google Docs.	• shapes, links, and page elements • Collaborate with others by sharing documents • Navigate through menus and toolbars.	• Demonstrate creativity and organization when formatting documents. • Value accuracy and neatness in digital document preparation. • Show respect for others' contributions when working collaboratively.	• Line spacing and paragraph formatting Working with lists and tables • Bulleted and numbered lists • Creating and formatting tables • Adjusting rows and columns Inserting Elements • Images, drawings, and shapes • Links and bookmarks • Headers, footers, and page numbers Sharing and collaboration • Sharing settings (view, comment, edit) • Working on the same document (with others in real time) • Using comments and suggestions	real-time editing and commenting. • Exploration Activity: In groups learners explore the toolbar options to discover additional formatting tools. • Project: Create a formatted class newsletter or report that includes text, lists, tables, and images.
Link to other subjects: <i>English: Writing, editing, and formatting essays or letters, Entrepreneurship: Creating professional project plans or proposals, Mathematics: Creating tables for data presentation.</i>				
Assessment criteria: <i>Learners explain the importance and features of Google Docs, create and format a document with proper layout, create a shared document with text, tables, and images.</i>				
Teaching and learning materials: <i>Computers with internet access, Google account, Projector</i>				

Topic Area: Computer System		Sub-topic area: Computer Hardware	
S4 ICT	Unit 2: Digital Logic Fundamental		Nº of periods: 14
Key Unit Competence: Design an optimized digital logic circuit implementing Boolean logic			
Learning Objectives		Content	Learning activities

Knowledge and understanding	Skills	Attitudes and values		
- Identify different types of number systems used in digital computing. - Explain the relationship between decimal, binary, octal, and hexadecimal systems. - Describe Boolean logic and its fundamental operators (AND, OR, NOT). - Explain how logic gates are interconnected to form logic circuits. - Discuss the process of simplifying Boolean expressions.	- Convert numbers from one system to another - Represent digital data using appropriate number systems. - Construct Boolean expressions from logical statements or circuit diagrams. - Simulate logic circuits using appropriate software tools. - Generate truth tables to verify Boolean expressions. - Apply optimization techniques to simplify logic circuits.	- Appreciate the importance of number systems and logic in computer operations. - Demonstrate accuracy and precision in performing conversions and simplifications. - Show creativity in experimenting with logic simulation tools. - Value the logic reasoning in computing processes.	Number systems - Types of number Systems (Decimal, Binary, Octal, and Hexadecimal) - Convert numbers Boolean algebra and logic gates Boolean logic and its operators (AND, OR, NOT) Logic Circuit Design - Logic simulation tools - Interconnect logic gates Constructing logical expressions - Boolean expressions - Truth Table Logic circuits optimization - Boolean rules - Simplification	- Individually learners convert different numbers from one system to another. - In groups learners identify a logical scenario and create its logical circuit. Activity: - In groups learners use logic simulation tools (e.g., Logisim, Tinkercad Circuits, Proteus) to create and test logic circuits. - In pairs learners construct Boolean expressions with their truth tables and simplify them - Groups exchange and test each other's logic circuits for functionality and accuracy.
Link to other subjects: <i>Mathematics: Base numbering systems, algebraic manipulation, and logical reasoning, Physics: Electronic circuits and digital signal representation.</i>				
Assessment criteria: <i>Accurately identify and describe all number systems; correctly convert numbers from one system to another; produce correct Boolean expressions and corresponding truth tables; design, connect, and simulate logic circuits accurately.</i>				
Teaching and learning materials: <i>Computers with logic simulation tools (Logisim, Tinker cad Circuits, or Proteus), a Projector for demonstrations</i>				

Topic Area: Computer Programming			Sub-topic area: Algorithm	
S4 ICT		Unit 3: Programming Algorithm		Nº of periods: 18
Key Unit Competence: Develop algorithmic solutions from problems using pseudocode and flowcharts.				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
• Explain the role, characteristics, and structure of an algorithm • Explain how data is stored in computer memory • Describe the importance of algorithms in problem-solving. • Explain the purpose of operators • Describe the input and output operations in algorithms. • Explain the concept and symbols used in flowcharts.	• Write an algorithm that solves a given problem • Identify variables, constants, and their data types within an algorithm. • Use reading and writing functions in algorithm • Write a Flowchart using symbols • Differentiate different data type used in computer algorithm • Convert pseudocode into flowcharts accurately and vice versa • Use statements (conditional and nested conditional)	• Demonstrate a willingness to follow algorithmic steps when solving computer problems. • Appreciate the importance of structured problem-solving. • Develop logical thinking and attention to detail when designing algorithms. • Value accuracy, organization, and clarity in	Introduction to algorithms: • Definition of algorithms • Characteristics of algorithm. • Importance of algorithms (in problem-solving) Variables and operations: • Definition of variable • Data type • Declaration • Operators (Arithmetic, logical, relational, increment and decrement) Input/output: • Initialization (variable and constant) • Read and write (statement) Algorithm representation: • Concept of flowchart • Flowchart symbols • Create basic flowcharts (from given algorithms or pseudocode) • Pseudocode to flowchart (Convert a pseudocode into a flowchart) Conditional statement • if	• In groups learners define algorithms, variables, and operators through class discussion. • In groups students explain the different data types and different operations on them. • Individually, students explain the use of the input and output functions. • In groups students provide appropriate steps to resolve a given problem, and draw a flowchart for it. • Individually, students write flowcharts from pseudocodes and vice versa.

• Explain the purpose and structure of conditional statements and loop statements.	- to represent decision-making. • Apply loop statements to represent repetitive tasks in algorithms.	algorithm representation.	• if else Nested conditional statements Nested if Loop statement • For loop • While loop	• Individually students write algorithms containing conditions, nested conditions and loops.
Link to other subjects: Mathematics: Logical reasoning, use of operators and sequence of operations.				
Assessment criteria: Students can derive a logical solution for a problem, represent it using a flowchart, write an algorithm, utilize variables and expressions in an algorithm correctly				
Teaching and learning materials: Computers, Projector				

Topic Area: Computer Programming			Sub-topic area: Programming Languages	
S4 ICT		Unit 4: Python programming I		Nº of periods: 17
Key Unit Competence: Create Python programs that involve decision making and operations on data.				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
•Explain the importance of programming in solving real life problems. •Explain how to install and run Python •Identify different data types •Explain the difference between variables and constants.	•Install and run Python •Write simple Python programs that declare variables and use appropriate data types. •Use input and output functions to interact with users.	•Value the use of programming in problem solving •Appreciate accuracy, organization, and logic in coding. •Demonstrate collaboration and knowledge	Introduction to Python •Python overview. •Python Installation Variables operation •Data types (int, float, str, bool). •Variable •Operators Input and output functions •Input function •Output function Conditional statements	•Learners in groups, explain the concept and importance of Python; give examples of where Python is used (e.g., web, AI, data science). •Individually, learners write simple programs using variables, data types, and operators •Learners use input() and print() functions to

• Explain the use of operators • Explain the use of input and output functions. • Explain the use of conditional statements • Explain the role of loops in programming	• Use different types of conditional statements in a program to make decisions. • Write a program using different types of loops	sharing with peers. • Value the role of programming in real-world problem-solving.	• If, if...else, if...elif...else Nested conditions • Nested if While loops statements • While loop For loops statements • For loop	create simple interactive scripts • In Groups, learners develop a project combining variables, conditions, and loops
Link to other subjects: <i>Mathematics: Logical reasoning, arithmetic operations, comparison operators, English: Writing and understanding syntax and keywords in English like commands</i>				
Assessment criteria: <i>Learners explain the use of variables, data types, operators correctly, and write a simple Python program using conditional and loop statements.</i>				
Teaching and learning materials: <i>Computers or laptops with Python installed (IDLE or online compiler), Projector</i>				

Topic Area: Programming			Sub-topic area: Web Design	
S4 ICT		Unit 5: HTML-Based Web development		Nº of periods: 18
Key Unit Competence: Develop a well-structured static website				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
•Explain the importance of HTML tags in web development •Identify different types of tags used	•Differentiate a webpage from a website •Create basic webpage using HTML tags.	•Value the importance of HTML as the foundation of web development.	Introduction to HTML • Definition of HTML • Role of HTML • HTML page structure (head and body,) Page structure tags	• Brainstorming: Discuss what makes up a webpage and explain the term HTML

to develop a web page • Describe the structure of an HTML document • Describe the basic structure of an HTML document • Identify the main content of a web page • Explain how to format text using appropriate tags • Explain the importance of using tables and forms on a web page	• Format text using appropriate HTML tags • Construct tables to present data clearly. • Design simple forms for user input. • Create hyperlinks and basic navigation links between web pages.	• Show creativity in designing webpages. • Develop accuracy and attention to detail when writing HTML code. • Demonstrate patience and persistence when solving coding errors. • Collaborate and share ideas responsibly when working on group web projects.	• <html>, <head>, <body> • <title>, <meta>, <link>, <!DOCTYPE> Text formatting tags Control how text appears: • Headings: <h1> to <h6> • Paragraphs: <p> • Emphasis: , <i>, <u>, • Marque List tags • Unordered List ; ordered list ; list Media tags • Embed images , • Audio <audio>, • Video: <video>, Table tags Structure tabular data: <table>, <tr>, <td>, <th>. Form elements tags Handle user input: <form>, <input>, <text area>, <button>, <select>, <option> Link and navigation tags (enable hyperlinks and navigation) <a>	• Students write HTML page and use the formatting features (bold, italic, superscript, subscript, strike through, heading, blockquote and text alignment and break tags) • Design a table in HTML and using <table>, table header<th>, table data<td>colspan and rowspan features. • Group activity: Students collaborate to design a mini personal webpage combining text, media, tables, and links.
Link to other subjects: English: Writing meaningful text content for web pages				
Assessment criteria: Learners correctly define HTML and explain its role, demonstrate understanding of HTML page structure and use of tags, write and run error-free HTML code, use tags appropriately to format text, embed media, and create forms. Create a simple, well-organized webpage that includes headings, images, lists, links, tables, and forms.				
Teaching and learning materials: Computers with text editors (Notepad, Notepad++, Visual Studio Code), a Projector for demonstrations.				

Topic Area: Programming			Sub-topic area: Web Design	
S4 ICT		Unit 6: Cascading Style Sheet		Nº of periods: 10
Key Unit Competence: Apply styles in web pages for better visual presentation and usability				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
• Explain concept of Cascading Style Sheets (CSS) and its role in web design. • Describe how CSS controls the appearance and layout of HTML elements. • Identify different text formatting properties • Explain how combining text styling and layout styling contributes to visually appealing web pages.	• Apply text formatting properties • Interpret the structure and syntax of CSS rules • Design a webpage section using proper font, spacing, and background properties. • Implement borders, padding, and layout adjustments effectively.	• Value the role of good layout and spacing • Demonstrate responsibility in following web design standards and best practices. • Show creativity when selecting colors, fonts, and backgrounds for a webpage.	Text formatting • Font styling • Text appearance • Spacing & layout • Effects Layout styling • Borders and padding elements (width, height, margin, padding, border, box-sizing:) • Background of elements (background-color, background-image, background-repeat, background-size, background-position) Styles with internal CSS Styles with external CSS	• Brainstorming: Learners discuss what makes a web page visually appealing and identify examples of good and poor text formatting from sample web pages. • Demonstration: The teacher demonstrates how to apply basic text formatting properties (font, color, alignment) using internal CSS in a live coding session. • Practice: Students write CSS rules to style given HTML text using font properties such as font-family, font-size, and font-style. • Learners create and link an external CSS file (and other types of CSS) to an existing

	Distinguish between internal and external CSS		Styles with inline CSS	HTML document, edit it, and observe changes.
Link to other subjects: <i>Mathematics: Use of measurements, proportions, and units to manage spacing and layout, English: readability and presentation of text content, Entrepreneurship: design attractive webpages for marketing</i>				
Assessment criteria: <i>Learners explain how spacing, borders, and backgrounds affect webpage design, differentiate internal from external CSS, write valid CSS code with appropriate syntax and linking methods, and design a webpage section using proper font, spacing, and background properties.</i>				
Teaching and learning materials: <i>Computers or laptops with internet access and a code editor (Visual Studio Code, Notepad++, or Sublime Text)., Web browsers (e.g., Chrome, Firefox, Edge) for previewing and testing webpages, Projector or smart board for teacher demonstrations</i>				

Topic Area: Database			Sub-topic area:	
S4 ICT		Unit 7: Database Design		Nº of periods: 16
Key Unit Competence: Design a normalized relational model using entity-relationship diagrams and appropriate keys				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
• Explain the use of a database • Explain different database access levels	• Distinguish between primary keys and foreign keys.	• Appreciate the importance of well-structured databases • Demonstrate accuracy and	Database concepts: •Definition, •Database access levels and users •Physical independence •Relational database model	•In groups, learners identify areas where database can be applied and learners design conceptual,

• Describe the concept of physical data independence. • Explain the difference between primary keys and foreign keys. • Explain different types of relationships • Describe normalization and its importance in database design. • Explain the Difference between first, second, and third normal Forms.	• Create tables with appropriate data types and constraints • Establish relationships between tables • Design Entity Relationship diagrams to represent database structures. • Normalize data structures up to third Normal Form.	consistency when designing a database. • Value data integrity, confidentiality, and reliability. • Collaborate with peers to model and solve database design problems. • Develop a positive attitude toward logical problem-solving in database design.	• Primary key • Foreign key Types of relationship (1-to-1, 1-to-Many, Many-to-Many) Entity relationship diagram • Entities • Attributes • Relationship • Representation of (E-R-D) Diagram Relational model Tables, columns, rows, primary keys, foreign keys, Data types Keys in relational model • Tables creation (with right data types and constraints: Primary key, foreign key) • Relation model representation (ERD to Relation model) Normalization • Normalization definition • Normalization role Types of Normal Forms First normal form Second Normal Form Third Normal Form	logical and physical levels of the database • Learners work on creating tables by defining structure, setting primary key, creating relationship among tables, use foreign tables and foreign keys, enforcing referential integrity rule, create query using design view • Individually, learners sit for exercises on Entity Relationship Diagram
Link to other subjects: Mathematics: Logical reasoning, relationships, and data organization.				
Assessment criteria: Explain and differentiate between various types of relationships, draw a correct and clear Entity Relationship Diagram (ERD), convert ERD into a relational model correctly, create database tables with correct data types, primary keys, and foreign keys, and normalize given data structures up to Third Normal Form.				

Teaching and learning materials: Computers or laptops with database software (e.g., MySQL, MS Access, SQL).

6.3. Senior Five

6.3.2. General competences of the subject at the end of Senior 5

- Create Google Sheets to solve real-life problems and organize information.
- Execute SQL commands for defining structure and manipulating data in database
- Develop Python programs using functions, lists, and object-oriented programming concepts
- Develop simple machine learning models to solve real-world problems
- Connect a web application to database using python
- Configure computer networks using Packet Tracer
- Explain the structure, functioning, and applications of Artificial Neural Networks

ICT units for Senior Five

Topic Area: Computer System			Sub-topic area: Computer Software	
S5 ICT		Unit 1: Google Sheet		Nº of periods: 10
Key Unit Competence: Create Google Sheets to solve real-life problems and organize information.				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
• Identify steps for logging into Google Workspace and opening Google Sheets. • Explain how to name and share a	• Access Google Sheets through Google Workspace. • Share a Google Sheet using correct sharing permissions.	• Appreciate the role of spreadsheets in organizing and analyzing data. • Demonstrate responsibility	Creating and sharing a Google Sheet link • Logging into Google workspace • Naming a sheet • Sharing permissions	• Practice creating a new Google Sheet, naming it, and sharing it with classmates. • Learners practice text formatting tools on given sample data, format cells using

Google Sheet using appropriate permissions. • Describe different text formatting options • Define arithmetic operations and basic functions • Outline types of charts and their uses in data presentation.	• Apply text formatting tools • Format cells by changing fill color, merging cells, and enabling text wrapping. • Apply arithmetic operations and functions • Customize the inserted charts to represent data visually.	when sharing documents • Exhibit teamwork and collaboration when working on shared Google Sheets. • Value digital tools as means for efficient data handling and presentation.	Text formatting • Bold, Italics, and Underline • Font size, color, alignment • Cell: fill color, cell merging, text wrapping Calculations: Arithmetic operations and functions (SUM, AVG, IF) Creating charts: • Types of charts • Formatting charts	colors, merging, and text wrapping, create various charts based on their data • Learners present their chart to the class, explaining how formatting enhances understanding.
Link to other subjects: <i>Mathematics: Application of arithmetic operations and averages.</i>				
Assessment criteria: <i>Log in and create a new Google Sheet, Application of bold, italics, underline, font color, size, and alignment, Use of spreadsheet formulas to compute results, shares correctly using “Viewer”, “Commenter”, and “Editor” permissions</i>				
Teaching and learning materials: Computers with internet access, Google Workspace accounts (for Google Sheets)				

Topic Area: Database			Sub-topic area:	
S5 ICT		Unit 2: Database queries		Nº of periods: 18
Key Unit Competence: Execute SQL commands for defining structure and manipulating data in database				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
• Explain the concept and purpose of SQL in database management. • Describe the categories of SQL languages and their functions. • Identify common SQL commands and their syntax. • Explain the importance of constraints • Describe the use of aggregate functions in data summarization. • Describe how to filter and sort data using SQL clauses. • Explain the importance of filtering and sorting data	• Install MySQL or similar database software. • Create a database using DDL commands. • Delete a database • Manipulate records using DML statements. • Apply constraints when designing database tables. • Use aggregate functions to analyze data. • Filter data using SQL clauses • Sort data using SQL clauses	• Value well-organized data using database management software • Appreciate the importance of data accuracy and integrity in databases. • Show curiosity in learning new SQL commands. • Work collaboratively during database projects • Value organization and documentation of queries for future reference. W	Introduction to SQL: • Database Queries • MySQL Installation • SQL Languages (DDL, DML, DCL) Data Definition Language (DDL) • Create, • Alter, • Drop Data Manipulation Language (DML) • Definition of DML • Operations (select, Insert, Delete, Update) • Inserting data into table • Selecting data from the table Updating and deleting data in a table • Update, delete SQL Constraints • Primary key, Not null, Foreign key Aggregate Functions Count, Sum, Avg, Max, Min	• Demonstration: Teacher demonstrates how to install MySQL and write basic SQL queries. • Discussion: Compare DDL and DML and discuss their respective uses. • Practice: Students practice creating and manipulating tables in MySQL. • In Groups: Learners design a simple database (e.g., Student Records or Inventory). • Learners perform data selection, insertion, updating, and deletion. • Problem Solving: Learners use aggregate functions and filtering clauses on given database records. • Project: Learners Create a sample

			Filtering and sorting data Where, And, Or, Order by (Ascending & descending)	database using DDL, insert data, and generate summary reports using aggregate functions.
Link to other subjects: <i>Mathematics: Use of logical and arithmetic operators in queries.</i>				
Assessment criteria: <i>Accurately Explain SQL, DDL, DML, and DCL, Successfully install and runs MySQL, Use CREATE, ALTER, DROP correctly, Insert, update, delete, and select data accurately, Implement primary key, not null, and foreign key constraints correctly, Apply COUNT, SUM, AVG, MAX, MIN effectively in queries and use WHERE, AND, OR, and ORDER BY clauses accurately.</i>				
Teaching and learning materials: <i>Computers with MySQL installed, Projector for demonstration</i>				

Topic Area: Computer Programming			Sub-topic area: Programming languages	
S5 ICT		Unit 3: Python programming II		Nº of periods: 15
Key Unit Competence: Develop Python programs using functions, lists, and object-oriented programming concepts				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
•Explain a function and its purpose in programming. •Describe how to create and call functions in Python. •Explain the concept of parameters, arguments, and return values.	•Write Python functions that perform specific tasks. •Use parameters and return values to exchange data between functions. •Apply scope rules effectively when	•Appreciate the importance of modular and reusable code. •Demonstrate logical thinking and problem-solving through structured programming.	Functions in Python • Definition of a function • Creating and calling a Function • Function parameters and return values • Local and global variables Advantages of using functions Arrays (Lists) in Python • Definition of lists • Importance of arrays (lists)	•Groups Discussion: Explore why functions are used in programming and how they simplify code. •Coding Exercise: Learners create a Python function that calculates the sum and average of a list of numbers.

• Explain the difference between local and global variables. • Discuss the advantages of using functions in program design. • Explain the importance of arrays in data storage and manipulation. • Describe how to create, access, and modify list elements. • Identify and use common list methods • Explain the concepts of object and class in Object-Oriented Programming. • Describe how to create classes and instantiate objects in Python. • Explain the difference between attributes and methods in OOP.	working with local and global variables. • Apply the creation and manipulation of Python lists to store, update, and retrieve data. • Use loops to iterate through lists efficiently. • Apply common list methods to manage data dynamically. • Design simple Python programs using functions and lists. • Construct classes and instantiate objects in Python. • Manipulate object attributes and methods appropriately.	• Value collaboration and teamwork in coding projects. • Show persistence and creativity when debugging and optimizing code. • Exhibit ethical behavior and integrity when sharing or reusing code.	• Creating and accessing lists • Adding, removing, and updating list elements • Iterating through lists using loops Common list methods (append, remove, sort, etc.) Working with arrays (list) • Adding, removing, and updating list elements • Iterating through lists using loops Common list methods (append, remove, sort, etc.) Introduction to Object-Oriented Programming (OOP) • Concept of object and class • Creating a class in Python Creating and using objects Attributes and methods	Use local and global variables in examples to illustrate scope. Group Work: • Design and test programs that use multiple functions and lists. Practice: • Learners create and manipulate lists by adding, removing, and sorting elements. Use loops to iterate through lists and display outputs. Project: • Build a simple contact book or student record system using lists and functions. • Learners discuss the benefits of OOP compared to procedural programming, creating a class with attributes and methods.
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Link to other subjects: <i>Mathematics: Logical reasoning, problem-solving, and algorithms, Physics: Simulation of real-world systems using OOP models, Entrepreneurship: Data management in business applications.</i>
Assessment criteria: <i>Define and differentiate key Python programming concepts (functions, lists, classes) correctly, write correct functional Python code using functions and lists, demonstrate proper use of parameters, return values, and variable scopes, manipulate list data effectively using loops and list methods, create Python classes and instantiate objects correctly, and apply attributes and methods in solving real-life problems.</i>
Teaching and learning materials: <i>Computers or laptops with Python installed (e.g., Thonny, IDLE, or Jupyter Notebook), Projector or smartboard for live coding demonstrations, Python code examples and sample programs.</i>

Topic Area: Artificial Intelligence			Sub-topic area: Emergence Technology	
S5 ICT		Unit 4: Machine Learning		Nº of periods: 12
Key Unit Competence: Develop simple machine learning models to solve real-world problems				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
- Describe key concepts of data, its types, sources, formats, and the main qualities of good data such as accuracy and reliability - Explain different methods of data acquisition, such as automated collection, web scraping, APIs, and	- Evaluate datasets for quality by assessing accuracy, validity, reliability, and timeliness. - Apply different methods to collect data from various sources and apply preprocessing techniques to clean and transform it for analysis.	- Cultivate a positive attitude toward continuous learning in data management and analysis. - Appreciate the importance of clean and reliable data for accurate analysis and decision-making, and develop a positive attitude toward maintaining	Data Fundamentals - Data, information, and knowledge - Data types and structures - Data formats (Data frame) - Data sources - Characteristics of quality data Data Collection, Preparation, and Cleaning - Methods of data acquisition - Data preprocessing techniques - Handling missing data, outliers, and duplicates	- Guided by the teacher, students investigate various data sources, including sensors, IoT devices, APIs, online repositories, and databases. And Discuss the advantages and challenges of each source. - In groups Students collect sample datasets from various sources like online

sensors, and describe common data preprocessing techniques • Explain how data can be organized in spreadsheets and databases, and identify tools for data visualization, including Excel, Python libraries, and other applications • Define Machine Learning (ML), explain its purpose, and identify its main categories: Supervised, Unsupervised, and Reinforcement Learning. • Explain the steps of an ML pipeline: data collection, preprocessing, model training, and evaluation.	• Effectively organize and manage data in spreadsheets and databases, and analyze graphs and charts to extract insights for informed decision-making. • Differentiate between Supervised, Unsupervised, and Reinforcement Learning algorithms, and apply basic ML concepts to simple examples or datasets • Apply basic ML tools to build and test simple models. Build, train, and test ML models using practical tools.	data quality and transparency. • Appreciate the role of accurate data analysis in informed decision-making • Develop curiosity about emerging AI and ML technologies • Appreciate the importance of following an ML workflow for successful model building. Appreciate the practical applications and impact of ML in solving real-world problems.	• Introduction to data ethics in collection and storage Data Analysis and Representation • Data organization in spreadsheets and databases • Data visualization techniques • Descriptive and inferential statistics Machine Learning Foundations • Definition and purpose of Machine Learning (ML) • Relationship between Artificial Intelligence, Machine Learning, and Deep Learning • Categories of ML algorithms • Real-world ML applications ML Workflow and Model Building • ML pipeline • Datasets • Splitting data into training, validation, and testing sets • Model evaluation metrics • Concept of overfitting and underfitting Tools for Machine Learning • Google Teachable Machine • Python-based notebooks	repositories, APIs, or sensors. • With guidance of Teacher, Students work in groups to organize sample datasets in spreadsheets or a simple database. • In group of students, research ML applications such as recommendation systems, voice recognition, and predictive analytics, and analyze their impact on daily life and industry. • In group students explore the three main ML categories Supervised Learning (regression and classification) Unsupervised Learning (clustering and pattern recognition) Reinforcement Learning (feedback-based learning) and discuss real-world examples of each type.
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				In groups, students Build and test a model to predict whether a fruit is an apple or an orange.
Link to other subjects: <i>Mathematics: Statistics, probability, and algebra.</i>				
Assessment criteria: <i>Explain key concepts of data, information, and knowledge, identify types and structures of data, demonstrate proper data cleaning and preprocessing techniques, apply statistical and visualization methods correctly, Design and test a basic ML model, analyze model outcomes, and interpret evaluation metrics.</i>				
Teaching and learning materials: <i>Computers, Internet connection, Spreadsheet software (Google Sheets, Excel), Scratch ML extensions, Notebooks, Projector for demonstrations</i>				

Topic Area: Computer programming			Sub-topic area: Web design	
S5 ICT		Unit 5: Web database connectivity		Nº of periods: 14
Key Unit Competence: Connect a web application to database using python				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
•Identify the components of a web application •Explain the difference between front-end and back-end •Explain the process of connecting the frontend to the	•Design an HTML form for user input. •Create a database using SQL commands. •Develop Python scripts to connect the web application to a database.	•Value the importance of Python in modern web development. •Demonstrate creativity and innovation in developing user-friendly web-based interfaces.	Python in web development • Components of web app (frontend, backend) • Front-end to back-end connection (Python codes for connecting) Frontend and backend design • Design HTML form • Design database Connecting to a database SQL and Python to link to a database Inserting and retrieving data • Inserting inputs into a database	•Brainstorming: Learners identify examples of popular web applications and discuss their frontend and backend components. •Practice: Learners design their own HTML forms, create corresponding SQL databases, and

backend using Python code. • Describe how HTML forms and databases are designed for data collection and storage. • Explain how Python interacts with SQL databases to perform CRUD operations.	• Implement data insertion, retrieval, updating, and deletion (CRUD) functions using Python. • Debug code to ensure successful frontend and backend communication.	• Collaborate effectively with peers when building and testing web applications. • Value the role of accuracy and consistency in programming and data management.	• Display stored data from a database Update and Delete data • Update data into database • Delete Data from database	write Python scripts to perform data insertion and retrieval. • Project: In groups, learners develop a mini web application (e.g., student registration system) that includes all CRUD operations.
Link to other subjects: <i>Mathematics: Logical thinking and data structure relationships</i>				
Assessment criteria: <i>Clearly explain frontend and backend roles, successfully connect and manipulate data, create a well-structured, labeled, and functional HTML form, develop Python scripts to connect the web application to a database, and implement data insertion, retrieval, updating, and deletion functions using Python.</i>				
Teaching and learning materials: <i>Computers with Python and MySQL/SQLite installed, Internet connection, Database tools (phpMyAdmin, etc.), HTML editor, Projector for demonstration</i>				

Topic Area: Networking			Sub-topic area:	
S5 ICT		Unit 6: Computer Networking		Nº of periods: 14
Key Unit Competence: Configure computer networks using Packet Tracer				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		

• Explain what a computer network is • Describe the importance and benefits of networking. • Explain the difference between wired and wireless network connections. • Identify different types of networks • Explain the functions of each type of network device • Describe basic network configuration concepts • Identify different network troubleshooting ways • Explain ping commands and their roles	• Distinguish between wired and wireless network connections. • Illustrate different network types using diagrams. • Set up network devices using Cisco Packet Tracer or similar tools. • Configure IP addresses and subnet masks • Test network connectivity using ping commands • Troubleshoot network connectivity errors	• Value the importance of communication and data sharing through networks. • Develop teamwork and collaboration when building and testing networks. • Demonstrate accuracy, responsibility and configuration tasks. • Value cybersecurity awareness and safe network practices.	Introduction to computer networks • Definition of a computer network • Importance and benefits of networking • Types of network connections (Wired, Wireless) Types of Networks • LAN, MAN, WAN, PAN (differences, examples, advantages) Network devices • Devices: Switch, Hub, Router, Modem, Access Point, Network Interface Card (NIC) • Functions of each one • Connecting network devices (use Packet Tracer) IP addresses Network configuration (in Packet Tracer) • MAC address, IP_s • Subnet mask and default Gate way • Configuration of devices Network troubleshooting • Testing connectivity (ping commands) • Verifying connection • Correcting errors	• Brainstorming: Define a network and discuss real-life examples (e.g., school Wi-Fi, company LAN). • Group work: Compare wired and wireless connections (advantages and disadvantages). • Practical activity using Packet Tracer • Connect network devices, configure IP addresses, subnet masks, and gateways, test connectivity using ping commands. • Troubleshooting task: Identify and fix misconfigurations in a simulated network.
Link to other subjects: <i>Physics: Transmission of electrical and radio signals, Mathematics: Binary numbering, IP addressing, and subnetting.</i>				

Assessment criteria: <i>Correctly define networking concepts and identify devices and network types, explain functions of devices and distinguish between network types, successfully connect, configure, and test devices in Packet Tracer, and identify and correct configuration or connectivity errors.</i>
Teaching and learning materials: <i>Computers with Cisco Packet Tracer installed, Network device models (switch, router, hub, cables).</i>

Topic Area: Emerging Technologies			Sub-topic area: AI	
S5 ICT		Unit 7: Artificial Neural Network		Nº of periods: 14
Key Unit Competence: Explain the structure, functioning, and applications of Artificial Neural Networks				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
• Explain the main concepts of ANN • Explain the importance of ANN in Artificial Intelligence • Identify network architecture components • Describe the training process of neural networks • Explain the difference between types of ANN: Deep Neural Networks,	• Construct and visualize a simple neural network model. • Apply training of a neural network using a Python library • Analyze the structure and performance of trained neural networks. • Apply ANN models to solve basic problems in image or	• Appreciate the role of ANN in solving complex real-world problems. • Demonstrate curiosity and critical thinking in exploring AI concepts. • Exhibit ethical consideration in the use of AI technologies. • Show teamwork and	Introduction to Neural Networks Concepts in ANN Importance of ANN in AI Network architecture: Input, Hidden and output layers Training of Neural Networks Types of ANN: Deep Neural Networks, Convolutional Neural Network, Convolutional Neural Networks Transfer of learning in ANN Application of ANN: Image recognition Speech recognition Sample Python program for ANN Ethics in Neural Networks	Group Discussion: • Define ANN and discuss its inspiration from the human brain. • Use visuals or diagrams to show neuron and layer structures. Demonstration: Explain the architecture (input, hidden, output) using interactive simulations Group Activity:

Convolutional Neural Networks and Recurrent Neural Networks. • Explain transfer learning and its use in ANN. • Outline key applications of ANN • Describe how bias can be introduced in neural network models and its ethical implications. •	speech recognition. • Apply ethical data collection and preprocessing techniques when training neural networks.	collaboration in coding and problem-solving activities. • Promote transparency and ethical reasoning in neural network decisions.		• Students research different types of ANN and present their findings • Analyze how ANN is used in image and speech recognition systems • In groups, learners discuss AI ethical use in Neural Networks
Link to other subjects: <i>Mathematics: Linear algebra, statistics, probability.</i>				
Assessment criteria: <i>Clearly explain ANN structure and functions, fully functional Python ANN with correct output,</i>				
Teaching and learning materials: <i>Computer with Python (Anaconda or Google Colab), Internet access</i>				

6.4. Senior Six

6.4.3. General competences of the subject at the end of Senior S6

- Design a shared secure google environment for collecting data from different people
- Develop interactive websites using web content management systems and AI tools
- Apply basic concepts of AI Fuzzy Logic to solve real-life problems
- Create a simple chatbot conversation using Python and Google Dialogflow
- Develop a simulated IoT system using the Arduino IDE.
- Use AI for enhancing a robotic system behaviour in a simulated environment

ICT units for Senior Six

Topic Area: Computer System			Sub-topic area: Computer Software	
S6 ICT		Unit 1: Google form		Nº of periods: 10
Key Unit Competence: Design a shared secure google environment for collecting data from different people				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
• List the elements constituting Google form interface with their uses. • Describe the process of creating a simple google form and sharing its link • Explain the different features of a quiz in a Google form and	• Use Google form application to create a form that receives user’s responses • Create a Quiz in Google form that provides prompt correction after submission. • Arrange exported data in tables and charts by using appropriate functions available	• Appreciate the design of the Google form interface and the form created in it. • Appreciate the use of google forms in collecting responses • Appreciate the quickness of correction in quizzes created in Google form	Introduction to Google Forms • Importance of Google Forms • Google Forms interface Creating a simple form • Steps to create a new form • Adding different question types (multiple choice, short answer, dropdown, checkboxes, etc.) • Adding more elements (titles, descriptions, and sections)	• In groups, learners brainstorm the importance of Google form and the use of its different interface elements. • Individually learners create forms having different features and explain the use of each feature • In pairs, Learners fill the forms shared to

how they are used. • Interpret google form responses in Excel sheets of charts. • Explain how to view and export Google Form responses	in Google form and Spreadsheet. • Analyse the relationship between google form fields and related spreadsheets columns.	• Sharing the link Designing Quizzes • Converting a form into a quiz • Assigning points and correct answers • Automatic grading and feedback Managing and analyzing responses • Viewing responses in Google Forms • Exporting data to Google Sheets • Simple charts and interpretation	them by their colleagues. • In groups, learners create a quiz that provides automatic grading and feedback. • In groups, learners export data from the forms, create charts and establish a relationship between the form fields and the generated Spreadsheet columns.
Link to other subjects: Mathematics (Histograms, cumulative frequency, probability modeling)			
Assessment criteria: Learners create Google forms and quizzes, export data and arrange it in tables and charts.			
Teaching and learning materials: Computers with internet connectivity, projector, access to gmail.			

Topic Area: Computer Programming	Sub-topic area: Web Design
S6 ICT	Unit 2: Website Development with AI
	Nº of periods: 18

Key Unit Competence: Develop interactive websites using web content management systems and AI tools				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
• Explain what a CMS is and its purpose in web development. • Identify the components of the WordPress dashboard. • Explain how to add text, images, videos, and menus. • Explain the purpose of plugins in WordPress. • Explain how AI tools can generate website layouts and content. • Identify basic HTML/CSS concepts used in AI-generated sites. • Describe the concept of responsive design. • Explain the role of SEO plugins • Recognize the importance of menu	• Compare different CMS platforms for suitability. • Navigate the dashboard and create pages and posts. • Customize themes to suit website requirements. • Organize menus for easy navigation. • Test interactivity features on the website. • Integrate AI-generated content into a website. • Integrate AI-generated elements for customization • Apply responsive themes layout for different devices. • Optimize pages and posts using SEO tools.	• Appreciate the efficiency and scalability provided by CMS. • Develop confidence in using WordPress. • Appreciate the role of design in user experience. • Develop attention to detail in presenting information online. • Appreciate the flexibility plugins provide. • Value innovation in web development. • Appreciate the combination of AI and manual design. • Value accessibility and user experience.	Web Content Management Systems • (WordPress, Wix, Joomla) • Benefits of CMS-based development WordPress • Installing WordPress (local) • WordPress dashboard • Creating pages and posts Themes and templates Installing and customizing themes Adding content and multimedia • Text, images, videos, and menus • Creating a basic homepage and about page Plugins and interactivity • Installing plugins (e.g., contact forms, sliders) • Enhancing user experience AI tools for website generation • AI in Web Development • How AI is used to design websites Website from Deep Site • Description of the the site, generate layout/content • Copy AI-generated HTML/CSS • Editing in WordPress • Integrating AI-generated elements • Site customization	• In groups, students compare WordPress, Wix, and Joomla, highlighting similarities and differences. • In group Students navigate the WordPress dashboard and identify the main menus and tools, customize site appearance (colors, fonts, and layout) using different themes. • With teacher guidance, students practice adding text, images, and embedded videos to posts and pages. • In pairs, students install and configure at least two plugins. • In groups, students use an AI tool to automatically

structure, links, and alt text • WordPress sites. • Explain the process of publishing a website online.	• Set up hosting for WordPress or HTML websites. • Use FileZilla, Panel, or online builders to upload a website.	• Develop attention to detail in content optimization. • Appreciate inclusivity in web design. • Develop responsibility in managing online resources.	Making websites responsive • Responsive themes (or customizing layout) SEO concepts • Title tags, meta descriptions, keywords • SEO plugins for WordPress (Yoast SEO). Improving navigation and accessibility • Menu structure, links, alt text for images Hosting the website • Domain name, web hosting providers • Hosting a WordPress website Uploading a website to a Web host • FileZilla (FTP), Panel, or online builders • Publishing the site and sharing the link	generate a homepage layout. • Students import and edit the generated content in WordPress, customizing colors and structure. • With teacher support, students preview their websites computer and tablet devices. • In groups Students install and configure the Yoast SEO plugin in WordPress. • With guidance from the teacher, students design a clear menu structure for easy navigation • Students simulate uploading a sample HTML file to a free hosting platform, connect to a hosting server using FileZilla or Panel.
Link to other subjects: <i>Mathematics:</i> For data analysis, chart and statistics, <i>Geography:</i> For environmental monitoring				
Assessment criteria: <i>Accurately evaluate practical skills, collaboration, creativity, and understanding, suitable final project</i>				
Teaching and learning materials: <i>Computers, projectors software, AI, Web development tools, Plugins and Extensions</i>				

Topic Area:	Sub-topic area:
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S5 ICT		Unit 3: AI with Fuzzy Logic		Nº of periods: 14
Key Unit Competence: Apply basic concepts of AI Fuzzy Logic to solve real-life problems				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
• Explain the difference between binary logic and fuzzy logic • Describe the evolution and concept of fuzzy logic. • Identify fuzzy sets membership functions. • Identify degrees of membership between 0 and 1. • Explain fuzzy rules • Describe how fuzzy inference systems work.	• Differentiate between binary logic and fuzzy logic. • Apply fuzzy logic operators to solve real-world decision problems. • Construct fuzzy sets and membership functions graphically. • Analyze linguistic variables. • Compare fuzzy logic operators with Boolean operators. • Develop fuzzy if-then rules for specific scenarios. • Simulate a basic fuzzy inference system using	• Value the flexibility and realism of fuzzy logic in handling uncertainty. • Demonstrate curiosity and creativity in exploring real-world fuzzy logic applications. • Value logical reasoning and approximate decision-making approaches. • Show teamwork and collaboration during group activities or simulations.	Fuzzy Logic • Binary Logic VS Fuzzy logic • Evolution of fuzzy logic Fuzzy Sets and membership functions • Fuzzy Sets • Membership Functions • Degrees of membership (values between 0 and 1) Fuzzy Logic Operators (AND, OR, NOT) • Fuzzy AND (minimum), OR (maximum), NOT (complement) • Comparison with Boolean operators Fuzzy rules and applications • Linguistic variable and values • Fuzzy if-then Rules Fuzzy logic in action • Fuzzy inference system (Concept) • Real-life use cases (Smart irrigation, Greenhouse control)	• Group Discussion: Compare binary and fuzzy logic through real-world examples • Learners demonstrate how membership functions represent degrees of truth. • Individually, learners create fuzzy sets and calculate membership degrees using examples. • Simulation: Use logic simulation tools or flowcharts to design a fuzzy inference model. analyze fuzzy logic applications in smart irrigation or greenhouse systems. • Group Work: Develop fuzzy if-then rules for a simple temperature control system. • Presentation: Learners present findings on real-

	appropriate software or flow diagrams. • Evaluate the effectiveness of fuzzy logic in practical systems like irrigation and automation.			world fuzzy logic systems.
Link to other subjects: <i>Mathematics: Set theory, probability, and functions, Physics: Measurement and control of physical quantities (temperature, humidity).</i>				
Assessment criteria: <i>Differentiate fuzzy and binary logic correctly, interpret fuzzy sets and membership functions accurately, apply fuzzy logic operators to solve defined problems, analyze fuzzy rules effectively, and explain the functioning of a fuzzy inference system.</i>				
Teaching and learning materials: <i>Computer with fuzzy logic simulation software (e.g., MATLAB Fuzzy Toolbox), Projector or smartboard for demonstrations.</i>				

Topic Area: Emerging Technologies			Sub-topic area: AI	
S6 ICT		Unit 4: Natural Language Processing		Nº of periods: 15
Key Unit Competence: Create a simple chatbot conversation using Python and Google Dialoglow				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
• Explain NLP and differentiate it with human language • Identify the three main components of a chatbot and	• Evaluate the role of NLP in human computer communication. • Demonstrate how a chatbot receives, processes, and responds to a simple user input.	• Demonstrate curiosity in discovering how NLP is used in developing today’s AI tools. • Demonstrate curiosity when exploring how	Introduction to NLP • Definition and overview • How computers understand human language. • Application of NLP and NLP tools Introduction to Chatbots	• In groups, students do a research on NLP and present their findings. Under the teacher’s guidance they summarize their findings. • In groups, students do a research on human

explain their uses. • Explain the purpose of setting up a Python environment before starting chatbot development. • Explain how a chatbot input, processing, and generation of responses in Python. • Explain how ChatterBot uses training data to improve chatbot responses • Explain the main components of Dialogflow • Differentiate human language, natural languages and programming languages. • Explain basic application of NLP including	• Install and configure Python environment for a chatbot • Write a basic Python script that simulates a chatbot • Train a created chatbot in Python chatterbot library • Apply debugging and logging to identify and resolve chatbot errors. • create a Dialogflow account and navigate its interface. • Apply basic NLP techniques such as tokenization and stemming using sample text data • Design a simple chatbot, translation, sentiments analyser, spell checker. • Demonstrate, using Python how a simple rule-based chatbot can be created.	chatbots understand and respond to language. • Appreciate the importance of properly setting up the Python environment. • Show respect to ethical principles when creating chatbot interactions. • Appreciate the importance of chatbot behavior in real-world applications. • Value continuous improvement and ethical responsibility in chatbot development. • Appreciate the value of using Dialogflow to build user-friendly • Appreciate the importance of proper setup in ensuring successful	• Components of a chatbot (interface, NLP engine, backend). • Role of NLP in chatbot Python environment preparation (for Chatbot Development) • Installing Python and IDE • Installing chatbot libraries (ChatterBot) Creating a simple chatbot in Python Training a chatbot using Python ChatterBot library • Training data concepts (corpus, conversations, and responses). • Using ChatterBot's training module. Improving Chatbot responses and handling errors • Handling unknown responses. • Adding more training data. • Logging and debugging chatbot errors. Introduction to Google Dialogflow • Components: Intents, entities, and responses. • Why use Dialogflow	language, natural language processing and computer language and do a presentation. • In groups learners perform tokenization, stemming, and lemmatization on a text. • In group students Define chatbots and list real-world examples. • With teachers guideline Build a simple bigram or trigram model using a short text corpus to predict the next word. • In group students discuss on how logic programming differs from procedural programming • With a guidance of teacher students Build a basic Q&A chatbot using Prolog facts and rules. • In group, students Write small Python scripts using variables, strings, and lists. • With guidance of teachers, students Compare manual results vs. NLTK outputs to understand library benefits. • In group students Build a simple Python chatbot
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chatbots, machine translation, spell checking	• Explain the concept of <i>next-word prediction</i> and how probability is applied in predicting text. • Apply list operations to store and manage tokens or words for chatbot or NLP tasks. • Install and import the NLTK library in Python.	chatbot development. • Develop a sense of responsibility in using chatbots,. • Appreciate the importance of data and context in making accurate language predictions. • Show curiosity and initiative in exploring open-source NLP tools.	• Dialogflow interface and creating an account. Creating a chatbot in Google Dialogflow (using free options) • Creating a new Dialogflow project. • Setting up intents (e.g., greetings, help, goodbye). • Adding training phrases and responses. • Training, testing and deploying a Dialogflow Chatbot • Enhancing chatbot with follow-up intents. • Testing chatbot performance. • Exporting or integrating chatbot with web interface.	using if-else statements for pattern matching.
Link to other subjects: <i>Mathematics (Probability & Statistics) for data analysis.</i>				
Assessment criteria: <i>Use Python and Google Dialogflow to create a simple chatbot</i>				
Teaching and learning materials: <i>Computers, projector, internet connection, Python, chatbot demos</i>				

Topic Area: Computer Programming			Sub-topic area:	
S6 ICT		Unit 5: Arduino Development		Nº of periods: 17
Key Unit Competence: Develop a simulated IoT system using Arduino IDE				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		

• Explain the importance and applications of IoT in daily life. • Describe the main components of an IoT system • Identify features of different Arduino boards • Identify and classify different IoT components in a given setup. • Describe the process of installing Arduino IDE and its components. • Explain how actuators respond to sensor inputs. • Describe the structure of an Arduino sketch • Identify local problems that can be solved using Arduino-based IoT systems.	• Identify examples of IoT systems in the surrounding • Distinguish digital and analog sensors. • Compare Arduino with other microcontrollers • Configure board and port settings for uploading code. • Connect different types of sensors (light, temperature, etc.) to the Arduino board. • Program the Arduino to trigger actuator responses based on sensor input. • Write Arduino sketches for reading sensors and controlling outputs.	• Show curiosity about how different IoT components interconnect • Show attentiveness and curiosity in exploring new technologies. • Value the importance of proper software setup in project success. • Work collaboratively to test and verify sensor outputs. • Appreciate systematic programming and documentation. • Demonstrate teamwork in identifying and discussing IoT components. • Develop a sense of innovation in solving problems using Arduino	IoT Components • Sensor/devices, • Connectivity, • Data Processing, • User Interface Arduino overview • Features of Arduino • Role of Arduino • Install Arduino IDE Sensor's connection to board and configuration • Library configuration Programming in Arduino • Structure of an Arduino sketch (setup () and loop ()) • Reading digital input using digital Read () Analog Sensors • Light sensors • Analog and Digital Input Actuators • Definition • Types of Actuators Controlling Servo motors • Using the Servo library • Build a mini gate control system	• In group learners identify the IoT components and explain their roles in an IoT system • Individually learners install Arduino and identify its features • In groups learners connect sensors to the board using simulated environments (or real ones if available) • In group students practice programming in Arduino and explain each programming code they use • In group learners and using simulations (or real environments where possible) connect a light sensor to an Arduino to read values and distinguish analog and digital inputs. • Learners explain different types of actuators and their functions in IoT systems. • In group, learners use the Servo library in Arduino to control a servo motor and build a simple mini gate control system. • Learners design an Arduino-based solution to a local problem by identifying the problem, selecting appropriate components, and demonstrating the expected system behavior.
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			Problem solving using Arduino	
Link to other subjects: <i>Physics: Understanding electricity, circuits, and how sensors and actuators work. Mathematics: Applying logic, algorithms learnt in Mathematics</i>				
Assessment criteria: <i>Ability to apply IoT and Arduino concepts by configuring hardware and software components, programming and testing sensors and actuators, and collaboratively designing a functional solution to a local problem. All these using simulated environment (or real one if available).</i>				
Teaching and learning materials: <i>Arduino Uno boards (or simulation software such as Tinkercad or Proteus), Sensors (light, temperature, motion, ultrasonic), Actuators (LEDs, buzzers, servo motors, DC motors), Computers or laptops with Arduino IDE installed, Internet access for downloading libraries and resources. Sensors and actuators are simulations (or real if available.)</i>				

Topic Area: Emerging Technologies			Sub-topic area: Robotics	
S6 ICT		Unit 6: Smart robots		Nº of periods: 14
Key Unit Competence: Use AI for enhancing a robotic system behaviour in a simulated environment				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
• Describe the difference between traditional robots and smart robots. • Describe the main components of an intelligent robot.	• Analyze features of traditional and smart robots. • Explain the functions of different	• Appreciate the importance of automation in modern industries. • Show curiosity and creativity in exploring	Smart Robots and Automation •Definition of automation and smart robots •Traditional VS smart robots •Role of Artificial Intelligence (in making robots “smart”)	Introduction: • Teacher-led discussion on daily life automation examples (e.g., washing machines, smart home devices).

• Explain how robots make decisions. • Describe simple decision-making techniques and how robots perceive the world. • Explain the process of designing AI-enhanced robotic systems. • Outline major applications of smart robots in various sectors • Explain how to design a simple smart robot using simulation tools. • Explain the role of humans in designing, supervising, and using AI-driven robots	robot components. • Demonstrate how sensors and actuators interact in a simple robotic system. • Apply IF–THEN rules to control a robotic system. • Design a basic AI-driven robot concept using simulation tool • Apply ethical guidelines when designing or interacting with smart robots.	AI-powered robotic systems. • Demonstrate teamwork and problem-solving during group projects. • Promote ethical use of robots in society to enhance, not replace, human work. • Respect for human dignity and the social implications of automation	• Components of an intelligent robot (Sensors, Processors/Controllers, actuators, AI software) Perception and decision-making • How robots perceive the world using sensors • Decision-making using simple algorithms or rules • Simple IF–THEN rules vs. learning models Applications of smart robots • Robots in medicine (surgery, elderly care) • Robots in agriculture (harvesting, spraying) • Smart home robots (vacuum, voice assistants) • Industrial automation (robot arms, inspection robots) Designing AI-enhanced robotic systems: - Smart Traffic Management Robot (use virtual environment) • Design process • Sketch or simulate logic flow of smart robot tasks • Programming the robot and integrating AI - Simulate a smart robot (using a simulated environment) • Design a simple smart robot (real or simulated using tools such as CoppeliaSim_Edu, OpenRAVE)	Presentation: • Explain traditional vs. smart robots with visual examples or videos. • Class Discussion: Explore AI’s role in autonomous vehicles or healthcare robots. Group work: • Identify and describe robot components using diagrams or physical kits. • Develop a flowchart for robot decision-making (using IF–THEN rules). Practical: • Build or simulate a simple robot model using microcontroller kits or software (e.g., Tinkercad, VEXcode, or Python simulation). • In groups student discuss different ethical guidelines when design or
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			AI Ethics with a robot • Humans' Role in Robotics and Automation • Ethical and social concerns • Responsible and fair use of AI in automation • Data privacy, security, and environmental responsibility	interacting with smart robots
Link to other subjects: <i>Physics: Motion, energy, and mechanical systems (motors, arms), Mathematics: Measurement, logic, and control algorithms.</i>				
Assessment criteria: <i>Identify robot components; demonstrate understanding of how sensors, processors, and actuators interact; simulate a simple AI-based robotic system or decision model.</i>				
Teaching and learning materials: <i>Computer and projector, simulation software (or robot kit), Diagrams showing robot architecture, Python or block-based programming tools (Scratch, Blockly, or VEXcode).</i>				

SUBJECTS AND WEEKLY TIME ALLOCATION FOR UPPER SECONDARY (S4-S6)